An Explainable Machine Learning System for Ten-Year Cardiovascular Mortality Risk Assessment

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I. Introduction

Cardiovascular risk prediction is useful only when the outcome, prediction horizon, source population, and reliability of estimated probabilities are specified. A classifier may discriminate between individuals with different outcomes while producing poorly calibrated probabilities. Our research therefore considers discrimination, calibration, and interpretability together. The specific question is whether eight baseline clinical and lifestyle variables can estimate ten-year cardiovascular mortality among US NHANES adults aged 40–79 without self-reported prior cardiovascular disease.

The study outcome is cardiovascular death within 120 months after examination. It does not include nonfatal myocardial infarction, nonfatal stroke, or prevalent cardiovascular disease. Predictions based on US survey data cannot be assumed valid for a hospital population or for Uzbekistan without external validation.

A. Research Outline and Contributions

The study is organized according to the Introduction–Methods–Results–Discussion (IMRAD) framework. Section II specifies the cohort, predictors, leakage-controlled development strategy, evaluation measures, and prototype architecture. Section III reports only the preliminary logistic-regression results currently available. Section IV interprets these findings, identifies unresolved methodological limitations, and distinguishes future experiments from completed analyses. The research contribution at this stage is a precisely defined ten-year fatal cardiovascular endpoint, a reproducible eight-variable baseline, and an explicit evaluation plan that treats calibration and explainability as distinct requirements.

II. Methods

A. Data and Cohort Definition

1) Data Sources and Outcome: The study combines public NHANES interview, examination, and laboratory data from 1999–2008 with the CDC 2019 public-use linked mortality files [1], [2]. Participants reporting prior congestive heart failure, coronary heart disease, angina, myocardial infarction, or stroke are excluded. Cardiovascular mortality is identified using underlying-cause categories 001 and 005 in the public mortality data, with follow-up restricted to 120 months.

TABLE I
Baseline Input Variables

Variable	Operational definition
Age	Years at examination (40–79)
Recorded sex category	NHANES RIAGENDR male/female category
Systolic blood pressure	Mean of available positive exam readings, mmHg
Body mass index	Recorded positive BMI, kg/m^2
Total cholesterol	Baseline laboratory value, mg/dL
HDL cholesterol	Baseline laboratory value, mg/dL
Current smoking	Yes/no/unknown from smoking questions
Self-reported diabetes	Yes/no/unknown; borderline and refused remain missing

2) Predictor Variables: The input schema contains exactly eight predictors (Table I). Mortality and follow-up fields are outcome labels only and are never model inputs. Unknown questionnaire responses are retained as missing rather than silently mapped to negative answers.

B. Model Development and Evaluation

C. Development and Temporal Evaluation

The development cohort comprises 8,902 adults from 1999–2006, with 274 ten-year cardiovascular deaths (3.08%). Five-fold cross-validation is designated for preprocessing, hyperparameter selection, and calibration. The later 2007–2008 cohort contains 2,941 adults and 87 events (2.96%). Because its outcomes were already inspected for the fixed logistic baseline, this temporal check must not be described as a completely untouched final test. A separate independent cohort would be needed for stronger confirmation.

D. Prediction Models

Logistic regression provides the already-trained interpretable baseline. Planned comparisons include random forest and one conservatively tuned boosted-tree algorithm (XGBoost or LightGBM). All imputation, encoding, scaling, model selection, and any calibration must be fitted using development folds only to avoid information leakage. More complex models are not presumed superior given the limited number of events.

E. Performance Measures

Discrimination is assessed using the area under the receiver operating characteristic curve (AUROC) and area un-

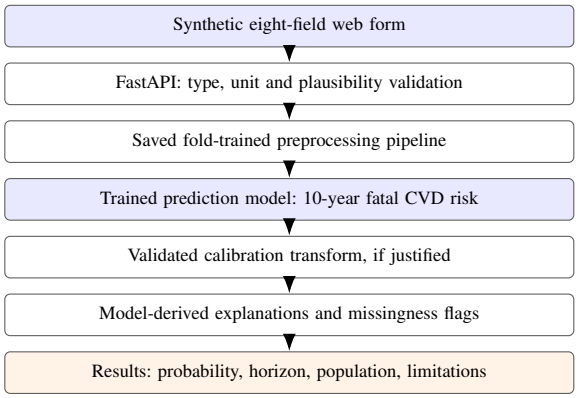


Fig. 1. Proposed explainable prediction prototype. Components beyond the baseline are planned, not reported as implemented.

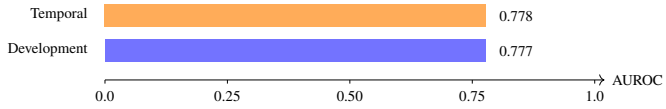


Fig. 2. Preliminary AUROC of the logistic-regression baseline (unweighted). Confidence intervals and additional metrics are given in Table II; similar AUROC does not establish external validity.

der the precision-recall curve (AUPRC), with AUPRC interpreted against event prevalence. Probability quality is assessed through calibration plots, calibration intercept and slope, and Brier score. Uncertainty intervals and event counts should accompany estimates. No clinical decision threshold has been established.

F. Proposed System Architecture

Fig. 1 presents the proposed research workflow. The demonstration follows seven stages: (1) a web form accepting eight defined fields; (2) FastAPI type, unit, and plausibility validation; (3) a persisted preprocessing pipeline; (4) a trained model producing a ten-year mortality probability; (5) an independently evaluated calibration transform if justified; (6) model-based explanations and missing-data context; and (7) a results page naming the outcome, horizon, population, and limitations. Synthetic example records, not real patient identifiers, are intended for the university demonstration.

SHAP may be used to characterize model contributions, but those contributions do not establish causal effects or benefits from modifying a risk factor. The interface must not invent a confidence score, risk-category cutoffs, or percentage-point contributions unsupported by the selected model and validation.

III. Results

Fig. 2 and Table II summarize the preliminary discrimination findings. Table II reproduces the reported preliminary logistic-regression findings from the project plan. They are unweighted, sample-level estimates, not clinical performance claims. The temporal AUPRC of 0.080 should be considered in relation to its 0.0296 event prevalence. The reported

TABLE II
Preliminary Logistic-Regression Performance

Metric	Development OOF	2007–08 check
AUROC	0.777	0.778
95% interval	0.749–0.803	0.738–0.819
AUPRC	0.124	0.080
95% interval	0.097–0.160	0.059–0.114
Brier score	0.0284	0.0281
Calibration intercept	-0.125	-0.223
Calibration slope	0.958	0.926

Brier-score intervals were 0.0255–0.0315 for development and 0.0231–0.0340 for the temporal check. Existing person-level bootstrap intervals do not account for the complex NHANES sampling design.

IV. Discussion

A. Interpretation of Preliminary Findings

The baseline AUROC values of 0.777 and 0.778 indicate similar ranking performance in the development and later-cycle samples. However, AUROC alone cannot determine whether individual predicted probabilities are reliable. The AUPRC values differ between the two samples, and both should be interpreted alongside the low event prevalence. The calibration intercepts and slopes suggest that probability assessment must remain central to subsequent work. These are descriptive findings rather than evidence of clinical utility.

B. Planned Experiments and Limitations

The next experiments will compare logistic regression with random forest and a boosted-tree model using development-only selection; evaluate fold-fitted imputation against complete-case analysis; test sigmoid or isotonic calibration within development data; assess explanation stability and wording; and explore age and recorded-sex subgroups only where event counts support meaningful estimates.

Several limitations require explicit treatment. There are only 87 events in the later cohort. Complete-case restrictions discard 54 development and 17 temporal events. NHANES sampling weights, strata, and clusters complicate population inference [7]. HDL laboratory methods changed across cycles. Public-use mortality files contain privacy-protecting substitutions. Finally, the model's fatal-outcome endpoint is not interchangeable with broader cardiovascular event risk scores or WHO charts [4].

C. Ethics and Clinical Translation

The current prototype is intended only for research and synthetic-case demonstration. It must not diagnose, triage, or recommend treatment. Clinical translation would require a defined intended use, independent local validation with known outcomes, clinician review, data governance, regulatory assessment, and supervised prospective evaluation [5], [6]. Entering real patient data would itself create privacy obligations. Reporting should follow the relevant TRIPOD+AI guidance [3].

REFERENCES

[1] Centers for Disease Control and Prevention, National Center for Health Statistics, "Public-use linked mortality files," CDC.

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[3] "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, 2024.

[4] World Health Organization, "WHO cardiovascular disease risk charts."

[5] World Health Organization, *Ethics and Governance of Artificial Intelligence for Health*.

[6] International Medical Device Regulators Forum, "Software as a Medical Device: Clinical Evaluation."

[7] Centers for Disease Control and Prevention, "NHANES analytic guidelines and weighting guidance."